

Senior Ontology - Smart Route & User Management API

Unit test specification: Find a route with pollution preference (4.0)

Scope note: Backend tests target server.py (FastAPI). DFD 4.1–4.5: same route input as 3.0; 4.2 adds preferred pollution via focus_pollutants on POST /scoreRoutes. Map / Pollution / Humidity data paths are mocked. Client route selection UI is out of scope.

Test suite — Process 4.0 Find a route with pollution preference

Test Case ID	TC-ROUTE-4.0
Python module	UnitTest_4_0_Route_With_Pollution_Preference.py
Priority	High
Data store	N/A (stateless scoring)

DFD sub-processes covered

- 4.1 Get route input — Same polyline + metrics as 3.0 (fixture).
- 4.2 Get pollution input — focus_pollutants e.g. ['pm2.5'] passed to score_routes; critic_weights boosts preferred pollutant.
- 4.3–4.5 Route list, selection, and detail — primarily client-side; server returns scored detail. Test: test_dfd_4_0_score_routes_with_focus_pollutants compares critic weights after baseline score_routes(None) vs score_routes(['pm2.5']).

Step	Action	Expected Result	Status
1	POST /scoreRoutes focus_pollutants None (baseline)	critic weights recorded (wrap)	Pass
2	POST /scoreRoutes focus_pollutants ['pm2.5']	pm2.5 weight higher; weights sum to 1	Pass
3	Mocked Map polyline + AQI + humidity	same fixture as 3.0	Pass

DFD / requirement	Unit test method
4.0 / 4.2 Preference in scoring	test_dfd_4_0_score_routes_with_focus_pollutants
Shared fixture	dfd_route_fixtures.score_routes_fixture

Coverage: run backend_tests/run_coverage.sh — UnitTest_4_0_Route_With_Pollution_Preference.py 100%. Generated 2026-04-06 10:53.